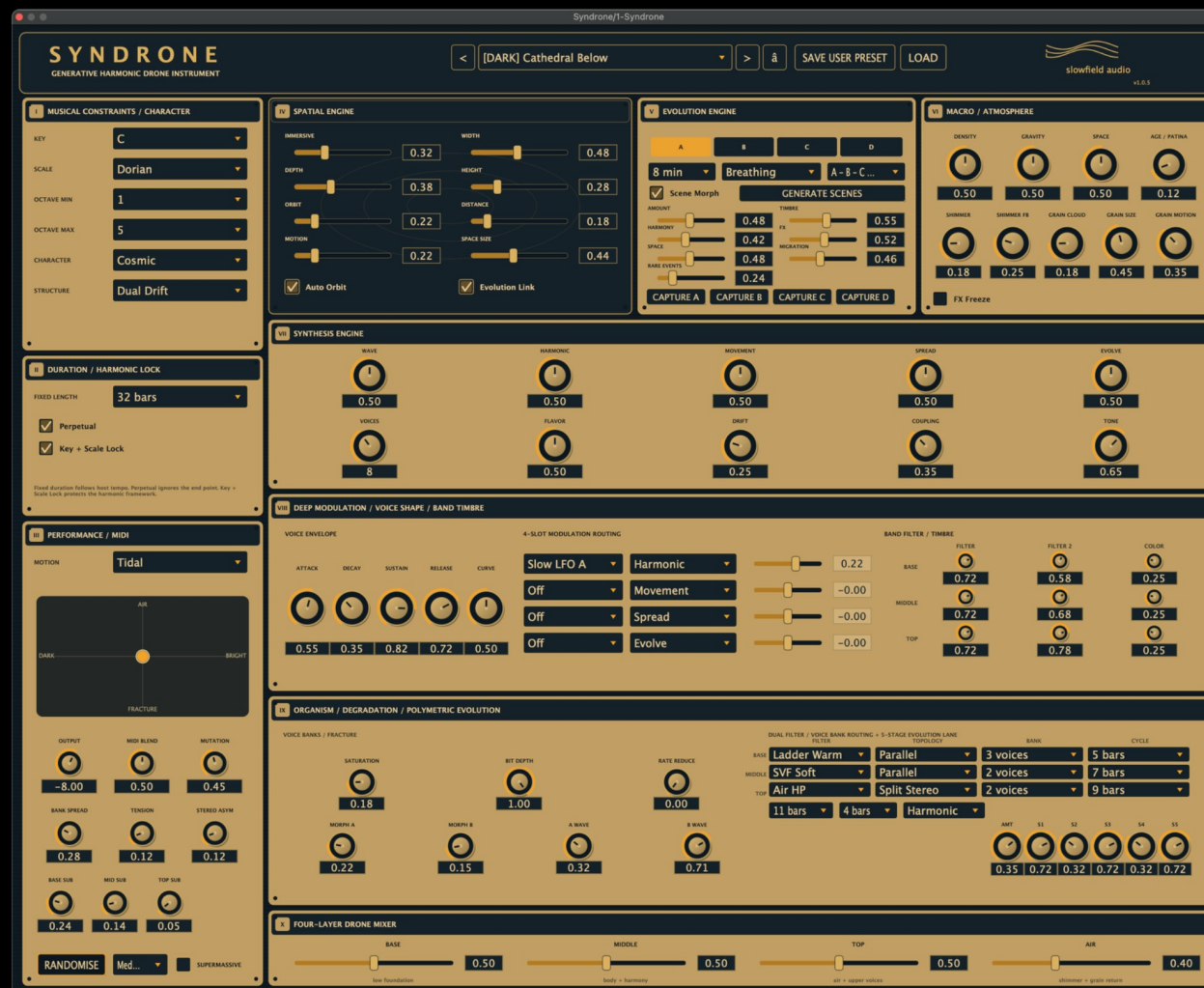


GENERATIVE HARMONIC DRONE INSTRUMENT

Expanded visual manual: synthesis, evolution, modulation, space, recipes and beta workflow



16 STRUCTURES

16 CHARACTERS

4 SCENES

4 MOD ROUTES

1-32 MIN EVOLUTION

A drone is a system, not a held note

Syndrone separates harmonic rules, sound identity, long-form change, spatial behaviour and layer balance.

1 HARMONIC WORLD

Key + Scale + octave constraints define the pitch vocabulary. Locking these rules lets autonomous movement remain musically related.

2 SOUND IDENTITY

Structure changes how the sound is generated. Character reshapes the resulting spectrum, density and personality.

3 TIME + SPACE

Evolution, scenes, modulation and spatial motion create change on several timescales - from pulses and LFOs to multi-minute form.

4 ATMOSPHERE + SCALE

Grain, shimmer, age/patina, Mass and Supermassive can move a patch from intimate and fragile to huge and cinematic.

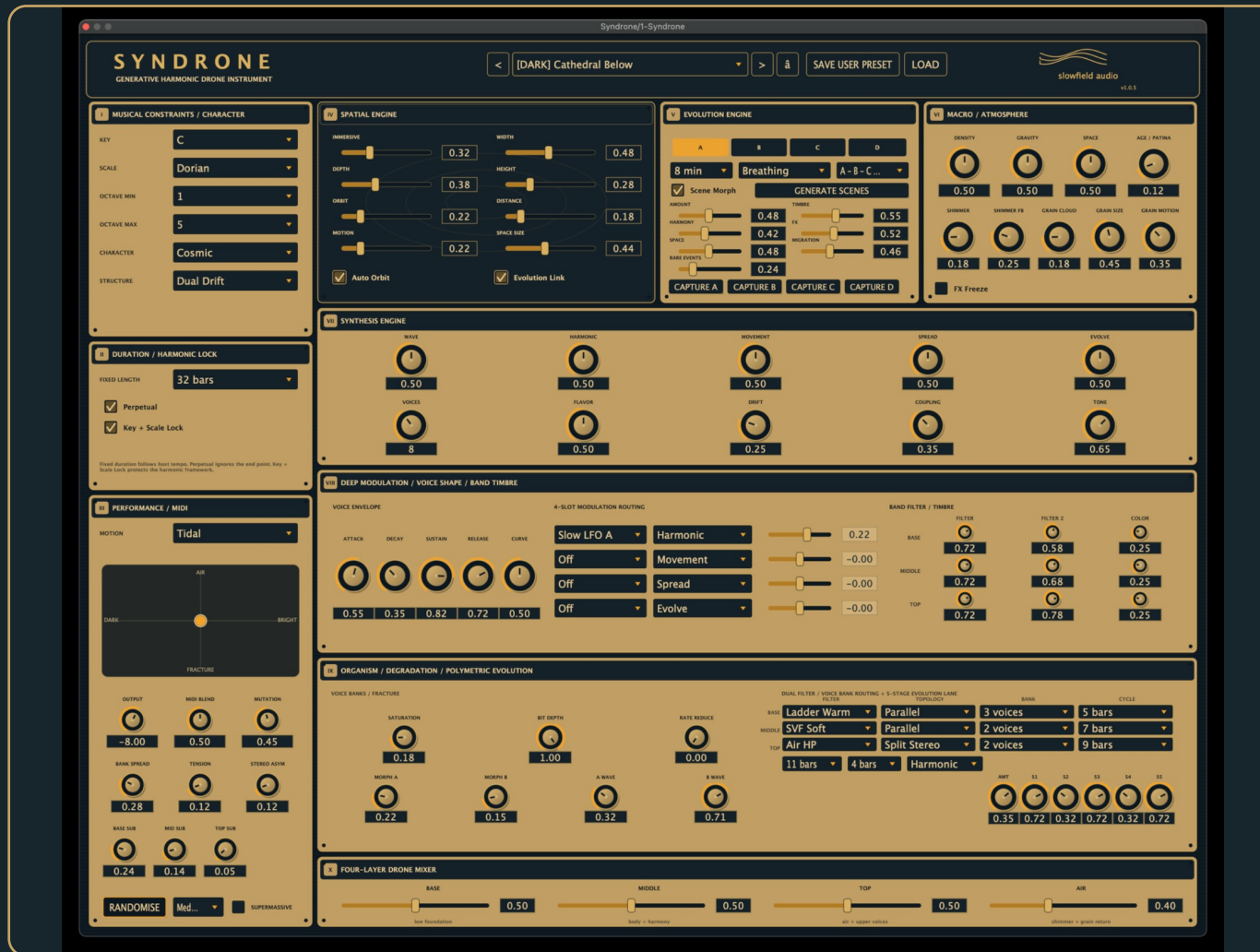
5 MIX + OUTPUT

Base, Middle, Top and Air are your final perspective controls. Balance them after the generative behaviour is working.

WORKFLOW PRINCIPLE > establish a musical constraint > choose a synthesis identity > add one slow movement > add space > balance layers.

The complete interface at a glance

Learn the geography first; detailed chapters follow.



LEFT / GENERATOR

Key, Scale, octave range, Structure, Character and core voice behaviour.

CENTRE / EVOLUTION

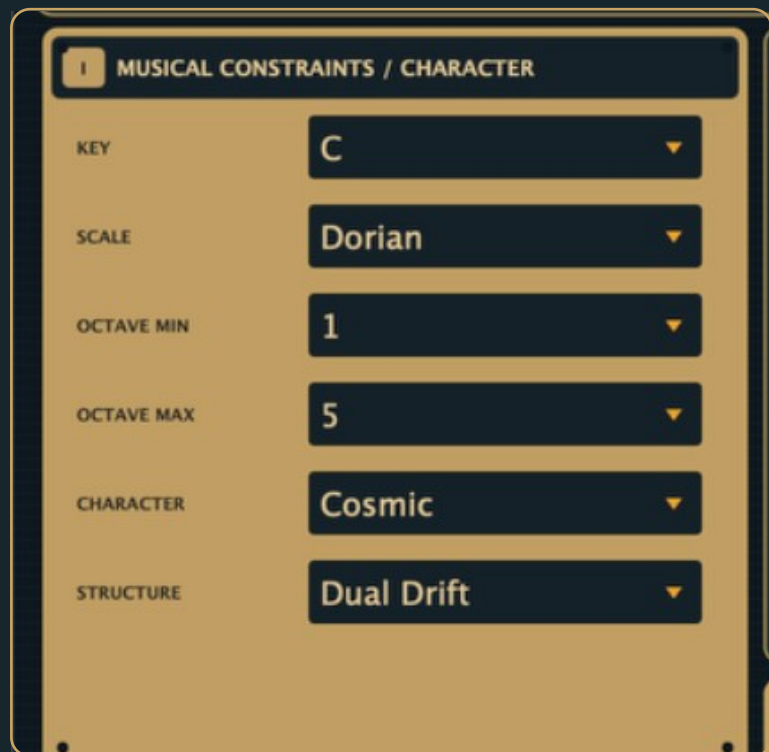
Scenes, scene path, evolution time/shape, rare events and related movement.

RIGHT + LOWER

Space/FX, modulation, Organism and four-layer mixer.

Key, Scale and octave range

Constrain the generator before increasing complexity.



KEY + SCALE

The Key establishes the tonal centre; Scale limits generated pitches to a selected pitch collection. For consonant ambient material, begin with familiar modal scales. For tension, try darker or less stable collections.

OCTAVE RANGE + LOCKING

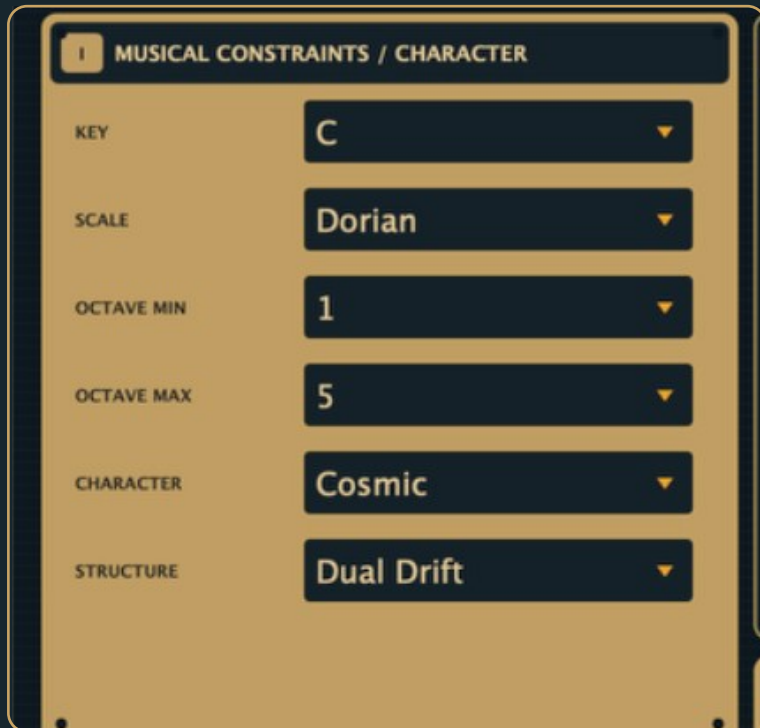
A narrow range produces focused, stable drones. A wider range increases vertical spread and the chance of dramatic register movement. Key/Scale locking is especially useful when Perpetual generation or randomisation is active.

TRY THIS

Create one patch in a two-octave range, save it, then widen the range without changing anything else. The comparison teaches you how much perceived complexity comes from register alone.

Structure is the synthesis architecture

It changes the way voices, harmonics and movement are assembled - not merely the waveform.



WHY STRUCTURE MATTERS

Choose Structure early. It establishes the raw behaviour that later controls - Character, evolution, modulation and FX - will animate.

TONAL / RESONANT

Useful starting families include Dual Drift, Subharmonic, Additive Resonator and Spectral Organ.

TEXTURAL / COMPLEX

Formant Choir, Phase Glass, Cluster Unison and Bowed Resonance are strong choices when timbral identity should be obvious.

LISTENING METHOD

Hold Character, scale and effects constant. Audition Structures one at a time. Listen for density, harmonic emphasis, transient/inharmonic content and how naturally each engine sustains.

Eight important Structure personalities

Descriptions below explain musical use rather than prescribing exact DSP implementation.

DUAL DRIFT

A broad starting point for slowly diverging voices. Use for warm beds and understated motion.

SUBHARMONIC

Emphasises lower harmonic relationships. Useful for weight, darkness and foundational drones.

ADDITIVE RESONATOR

Weighted harmonic/inharmonic partials suggest bells, metal and resonant bodies.

FORMANT CHOIR

Formant-weighted colour produces vocal, choral and vowel-like movement.

PHASE GLASS

Bright, crystalline spectra. Excellent with long scenes, shimmer and restrained modulation.

CLUSTER UNISON

Dense micro-detuned material for unstable clouds, tension and very large atmospheres.

BOWED RESONANCE

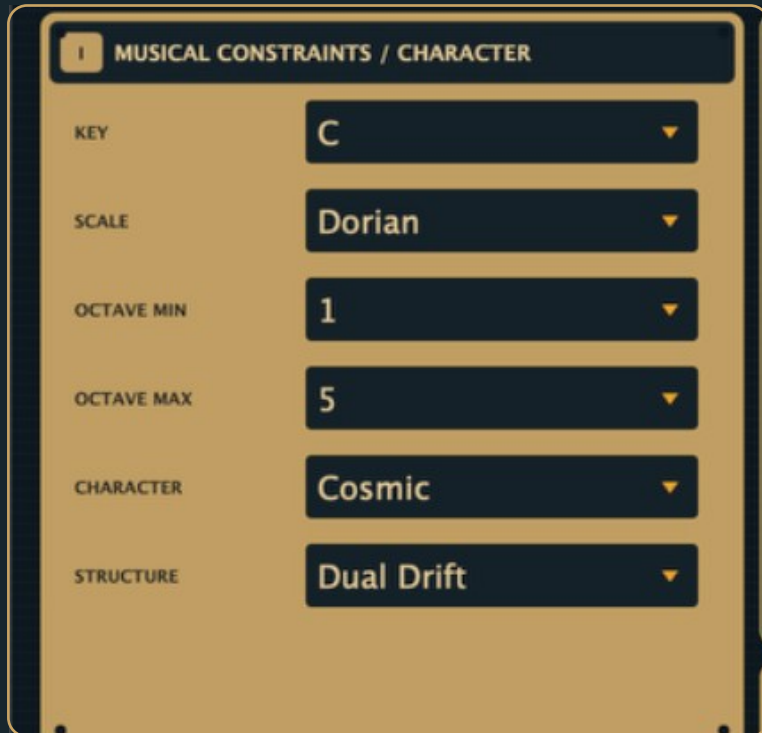
Resonant, body-like behaviour suited to organic, string-like and frictional textures.

SPECTRAL ORGAN

Structured harmonic stacks produce sustained, architectural and organ-like material.

Character reshapes the personality

Treat Character as a macro voicing decision after choosing Structure.



WHAT TO LISTEN FOR

Character can alter perceived harmonic weighting, movement, spread, tension and tonal personality. It is most revealing when effects are initially kept modest.

ORGANIC SIDE

Warm, Hollow and Fragile are useful starting points when the drone should breathe, recede or feel imperfect.

INTENSE SIDE

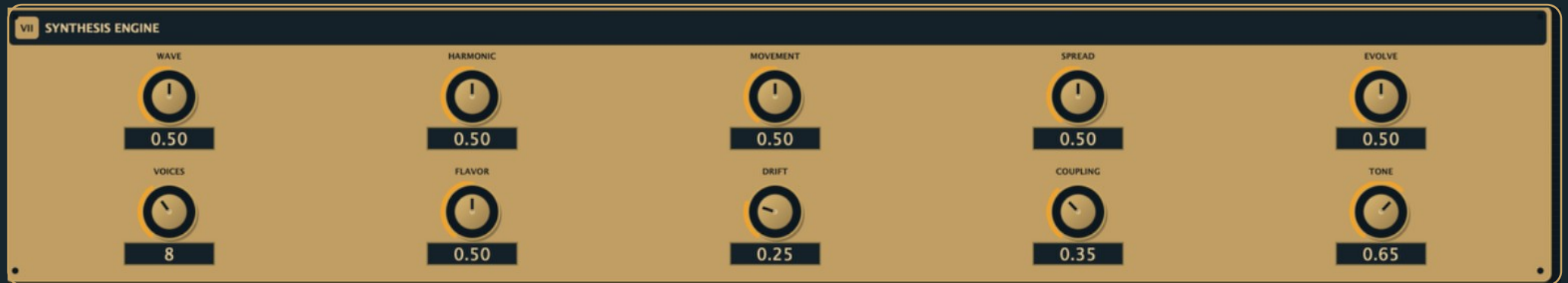
Dense, Mechanical, Monolithic, Radiant and Alien can push a Structure toward weight, brightness, rigidity or strangeness.

PAIRING EXPERIMENT

Try one Structure with three very different Characters. Then reverse the test: hold Character constant and change Structure. This quickly separates architecture from voicing.

Density, movement, tone and harmonic focus

These controls determine how populated and animated the core sound feels.



VOICE COUNT / DENSITY

Fewer voices expose individual movement. More voices create mass but can blur harmonic detail and consume more CPU.

MOVEMENT / SPREAD

Small amounts preserve a stable centre. Larger values create instability, width and beating.

TONE / HARMONIC

Use these before EQ-like corrective moves. They are useful modulation destinations because slow changes read as timbral evolution.

GAIN STAGING

Large voice counts, Mass, Supermassive and atmospheric FX can accumulate energy. Build the patch at conservative levels, then use the mixer/output stage to restore perspective.

Fixed forms versus Perpetual

Decide whether Syndrone should make a bounded phrase or an effectively endless environment

16 / 32 / 64 BAR FORMS

Use fixed lengths when the drone needs to fit an arrangement, cue or performance section. They are also easier to audition because the generative arc has a repeatable horizon.

PERPETUAL

Use Perpetual for installations, meditation, sound beds and exploratory sessions. Harmonic constraints and restrained modulation become more important because the patch may run for a long time.

PRACTICAL STRATEGY

Build and debug a patch using a fixed form first. Once the tonal balance, scene contrast and modulation are convincing, switch to Perpetual and listen for 10-20 minutes. Long listening reveals cycles, CPU behaviour, runaway density and repetition that short previews hide.

TIP > 'Endless' becomes interesting when several independent timescales coexist: slow scene travel, medium spatial motion and subtle faster modulation.

Scenes are snapshots of musical states

Use them to create intentional large-scale contrast.



CAPTURE

Create a useful state, then capture it to A, B, C or D. Think of each scene as a destination rather than a preset.

MORPH

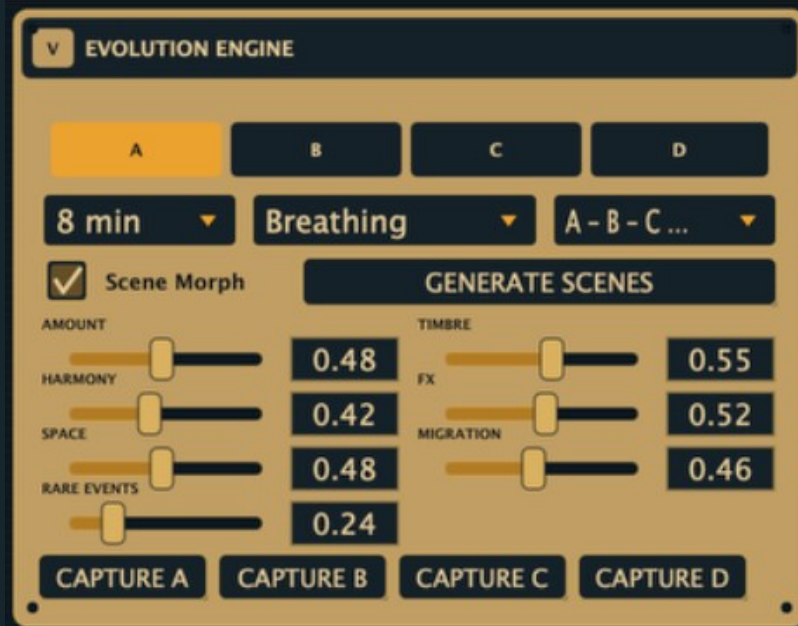
Scene Morph moves between states. Small differences create organic continuity; dramatic differences can make the drone feel like a composition with sections.

TWO-SCENE METHOD

A: dark, narrow, low and dry. B: brighter, wider, more animated and slightly wetter. Morph slowly between A and B before introducing C and D. This makes it easier to hear what the scene system is contributing.

Movement measured in minutes

Evolution turns snapshots into form.



EVOLUTION TIME

1-32 minute periods support everything from obvious movement to almost imperceptible drift. Start around 8 minutes for learning.

AMOUNT

Higher is not automatically better. Moderate amounts preserve the identity of a strong starting patch while still allowing change.

SHAPES

Rise/Fall suggests directional movement. Breathing implies recurring expansion/contraction. Wander is less goal-directed. Tide suggests broad cyclical movement. Bloom opens outward; Collapse moves toward contraction. Generative is useful when less predictable long-form behaviour is wanted.

Rare Events + Evolution Link

Controlled surprise is more effective than constant surprise.

RARE EVENTS

Use Rare Events as punctuation. At low levels they can make a long session feel alive. Too much can obscure the identity of scenes and make causality difficult to hear.

EVOLUTION LINK

Linking spatial behaviour to the evolving composition can make width, position or movement feel structurally related to the musical arc rather than like an independent effect.

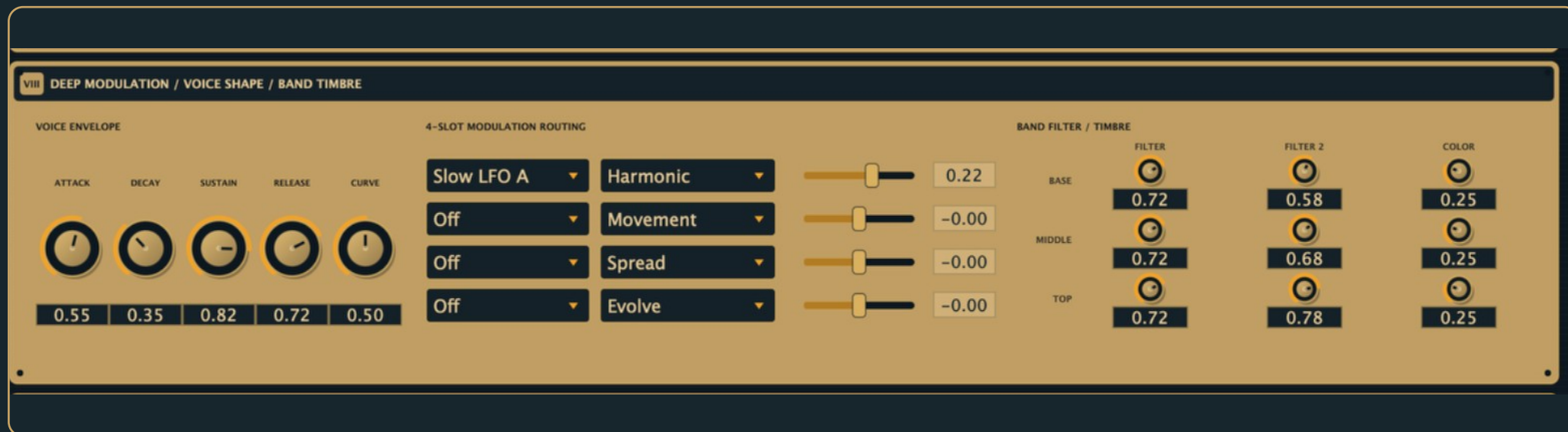
DEBUGGING A BUSY PATCH

Temporarily disable Rare Events, reduce Evolution Amount and use only two scenes. If the patch becomes coherent, reintroduce one source of unpredictability at a time. This is faster than randomly turning controls down.

HOT TIP > If a drone sounds impressive for 30 seconds but tiring after five minutes, reduce event density before reducing tonal richness.

SOURCE -> DESTINATION -> AMOUNT

Each of four slots can make a parameter move without manual intervention.



SOURCE

The changing control signal: LFO, random process, evolution state, pulse or host-related phase.

DESTINATION

The parameter being moved: tone, harmonic balance, orbit, age, filters, FX, layer levels and more.

AMOUNT

How strongly the source affects the destination. Start small; deep modulation is easier to add than to untangle.

A slow LFO used as a control signal is a standard synthesis technique; it can smoothly move another parameter such as filter cutoff rather than being heard directly as audio.

Predictable and synchronised sources

Use these when you want recognisable motion or host relationship.

SLOW LFO A-F

Smooth periodic movement. Excellent for orbit, width, tone, filters and effect depth. Different rates can create long composite cycles.

HYPER LFO

Use when you need a more assertive or complex cyclic modulator. Keep Amount conservative until its movement is understood.

ENVELOPE

A performance-related contour source. Useful when the patch is played from MIDI and modulation should respond to articulation.

STEP LANE

Discrete stepped movement. Useful for patterned changes that should be more obvious than an LFO.

HOST TEMPO PHASE

Relates movement to host timing. Useful when a drone must breathe with a DAW project rather than free-run.

SCENE / EVOLUTION

Scene Morph and Evolution can themselves drive destinations, linking local parameter change to the large-scale form.

Random, chaotic and probability movement

These are central to generative patches because repetition becomes less obvious.

RANDOM WALK

Successive changes wander from the previous state. Good for natural drift.

SMOOTH RANDOM

Random targets with softened transitions. Useful for width, tone and spatial motion.

SAMPLE & HOLD

Discrete random values held between updates. Good for stepped unpredictability.

SAMPLE & GLIDE

Random targets with transitions between them - less abrupt than sample-and-hold.

BROWNIAN WALK

A particularly organic wandering source. Excellent at low Amount into Tone, Orbit or Age.

CHAOS

Use for irregular movement with stronger unpredictability. Best introduced gradually.

BIPOLAR RANDOM

Moves on both sides of a centre point, useful when equal positive/negative deviation is desired.

PROBABILITY PULSE

Events occur probabilistically rather than every cycle. Useful for occasional filter, age or FX changes.

Euclidean Pulse and probability

Add structure without turning Svndrone into a conventional sequencer.

EUCLIDEAN PULSE

A Euclidean pattern distributes a chosen number of events across a cycle as evenly as possible. In a drone, use it to create recurring but spacious changes to filters, movement or effect parameters.

PROBABILITY PULSE

Probability can make some expected events occur and others disappear. This reduces obvious repetition and is useful for long-running patches.

RECIPE

Host Tempo Phase -> Movement (small amount). Euclidean Pulse -> Top Filter (moderate). Probability Pulse -> Age (small). Keep the synthesis itself sustained; let modulation provide the rhythmic implication.

TIP > Rhythmic modulation is most effective when only one or two destinations are strongly pulsed. Leave the rest of the drone continuous.

Where should modulation go?

Choose destinations by musical role.

TIMBRE

Tone, Harmonic and filters: ideal for slow spectral movement. Start here for subtle life.

SPACE

Orbit, Width and related spatial controls: movement can be audible without altering pitch or harmony.

WEATHERING

Age/Patina is effective for gradual deterioration or time-worn motion.

FX

Grain Size, shimmer feedback and related effects can make atmosphere itself evolve.

MASS

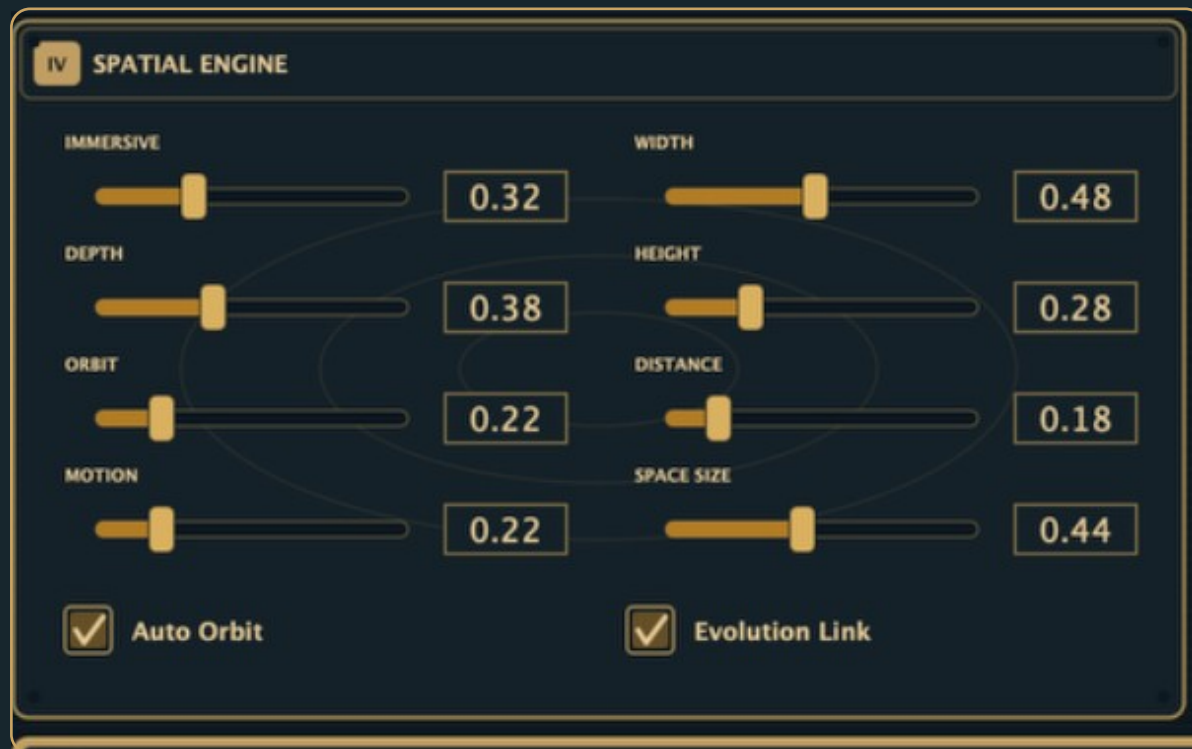
Modulating scale/density parameters can create large structural swells. Use conservative amounts.

MIXER

Layer levels can reshape the perceived spectrum without changing the synthesis engine.

Width, Depth, Height, Orbit and Distance

Treat space as part of the composition.



STATIC SPACE

Set Width, Depth, Height and Distance to establish the perceived environment before enabling automatic motion.

ORBIT + AUTO MOVEMENT

Orbit provides a strong sense of motion. Slow rates are usually more convincing for ambient work than constantly obvious movement.

DESIGN IDEA

Scene A: intimate and close. Scene B: wider and more distant. Link evolution to spatial change. The same harmonic material can then feel as if it moves from a small room into a huge environment.

Shimmer, grain and degradation

Use effects to extend the synthesis, not hide it.



SHIMMER

Adds bright, elevated ambience. Low settings can provide air; higher feedback can dominate the harmonic field. Watch accumulated energy.

GRAIN CLOUD / GRAIN SIZE

Granular atmosphere can diffuse edges and create texture. Modulating Grain Size slowly can make the cloud itself evolve.

AGE / PATINA

Age/Patina is especially useful as a slow modulation destination. Rather than setting a permanently degraded tone, let the sound weather gradually across a scene or long evolution cycle.

Scale without simply turning up the gain

Density and perceived size are compositional parameters

MASS

Use Mass to increase perceived scale, bank density and body. It can make a sparse patch feel architectural, but it also increases the need for careful layer balance.

SUPERMASSIVE

Pushes the impression of density and body further. Use it for monolithic or cinematic patches, then check the output level and low-frequency build-up.

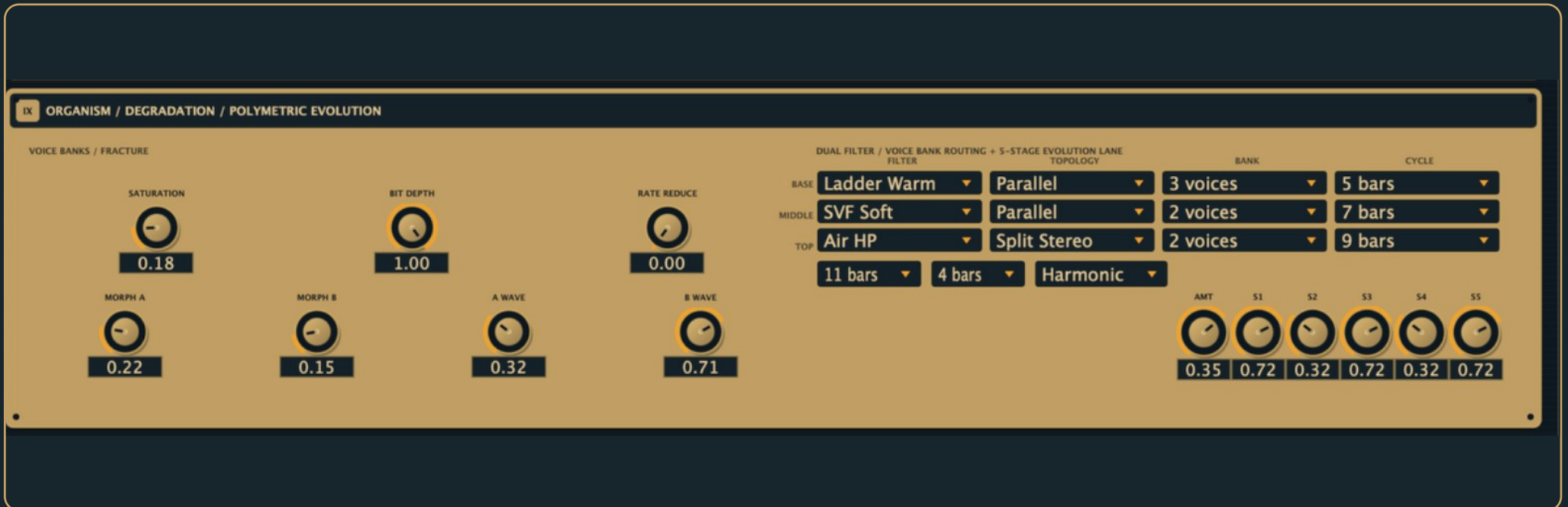
A/B METHOD

Save the patch. Reduce Mass/Supermassive to minimum and listen. Restore them gradually. If the patch only works at extreme density, simplify modulation or layer balance so the underlying synthesis remains interesting.

HOT TIP > Huge sounds often benefit from less shimmer, less Air and fewer simultaneous high-energy modulation routes.

Fracture, organic instability and micro-change

The Organism area is best approached as controlled complexity.



WHEN TO ADD IT

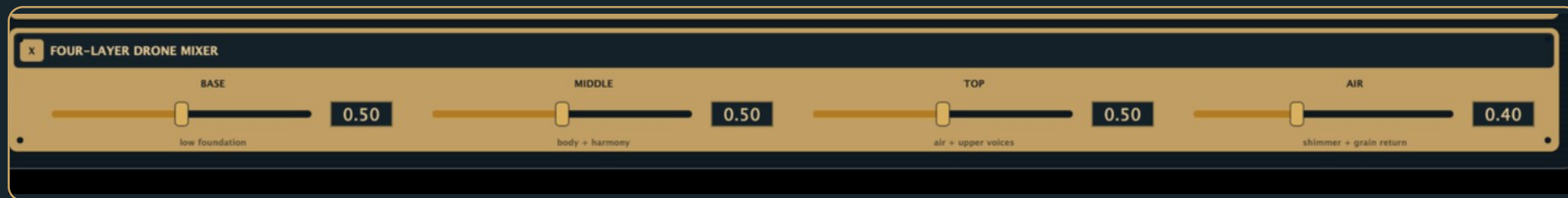
After the harmonic world and Structure already work. This lets you hear whether Organism adds life or simply masks a weak core patch.

HOW MUCH?

Start low. Increase until you clearly hear extra behaviour, then back off slightly. Generative sound often feels richer when the mechanism is not immediately obvious.

Base, Middle, Top and Air

The final mix is where an enormous patch becomes readable.



BASE

Foundation and weight. Too much can make Mass-heavy patches cloudy.

MIDDLE

Body and intelligibility. Often the anchor of a balanced drone.

TOP

Definition and upper detail. Useful when a dark Structure needs articulation.

AIR

Highest spectral impression. Reduce if shimmer or bright Structures become tiring.

MIX LAST

Do not use the mixer to compensate for every synthesis problem. First correct Structure, density and FX. Then use the four layers to establish perspective and headroom.

Explore without losing the patch

Syndrome includes 52 contrasting showcase and Starter patches.

PRESETS

Use Starter patches to learn one concept at a time. Showcase presets are better for hearing how multiple systems interact. Save a copy before major edits.

RANDOMISE

Gentle and Medium randomisation are often more productive when you already like a patch. Wild is better treated as a discovery tool rather than a finishing step.

SAFE EXPLORATION LOOP

1. Save current state. 2. Randomise. 3. Listen for one interesting change. 4. Reduce anything excessive. 5. Save as a new preset. 6. Compare against the original. This turns randomness into curation.

TIP > Randomise one creative stage at a time; do not judge a complex random patch before checking level, density and effects.

Three accessible starting patches

01 WARM BED

C Dorian
Dual Drift + Warm
4-8 voices
Narrow-medium octave range
8 min Wander
Moderate Width
Low Shimmer
Slow LFO -> Tone

Listen for: steady warmth with barely visible internal movement.

host

02 HOLLOW MEMORY

A minor / modal
Resonant Structure + Hollow
Low-moderate density
8-16 min Breathing
Age/Patina modest
Smooth Random -> Width

Listen for: faded, imperfect motion without obvious rhythmic modulation.

03 CHOIR CLOUD

D Dorian
Formant Choir
Warm or Hollow
Two scenes
Scene B wider/brighter
Light Grain Cloud
Brownian -> Orbit

Listen for: vowel-like colour moving through space.

VARIATION EXERCISE

For each recipe, save three versions: DRY, EVOLVING and SPATIAL. This creates a personal reference library and makes it much easier to understand which Syndrone subsystem is producing the result.

Three deeper generative patches

04 GLASS CATHEDRAL

Lydian
Phase Glass + Radiant
A/B scene morph
16 min Bloom
Shimmer + Grain Cloud
Brownian -> Orbit
Evolution -> Harmonic

Keep the high end controlled; let brightness arrive slowly.

and

05 TIDAL ORGAN

Spectral Organ
Two-octave range
8-16 min Tide
Scene A dry/near
Scene B wet/distant
Slow LFO -> Width
Evolution Link ON

Aim for broad recurring expansion without a short obvious loop.

06 BROKEN RADIO SKY

Fragile Character
Moderate Age
Sample & Glide -> Tone
Probability Pulse -> Age
Low Grain Cloud
Sparse Rare Events

Aim for intermittent weathering rather than constant degradation.

INTERMEDIATE RULE

Use only one strongly audible modulation route. The other routes should operate below the threshold where the listener can easily identify the mechanism.

Three self-evolving systems

07 ALIEN MONOLITH

Locrian / Phrygian
Cluster Unison + Alien
Mass + Supermassive
16 min Generative
Probability -> Top Filter
Brownian -> Age
Slow LFO -> Orbit

Keep mixer Air restrained.

scale

08 GENERATIVE PULSE FIELD

Sustained Structure
Host Tempo Phase -> Movement
Euclidean -> Filter
Probability -> FX depth
8 min Wander
Rare Events low

The pulse should imply rhythm while the core remains a drone.

09 FOUR-SCENE JOURNEY

A dark/close
B bright/wide
C dense/aged
D sparse/distant
16-32 min evolution
Scene Morph -> Width
Evolution -> Harmonic

Use this as a complete evolving cue.

ADVANCED RULE

When several modulators are active, give each one a different role: one for timbre, one for space, one for occasional events and one for macro form. Avoid four routes all fighting over the same audible dimension.

When a patch stops working musically

Reduce complexity methodically instead of starting over.

TOO BUSY

Disable Rare Events; reduce modulation Amount; use two scenes; narrow octave range.

TOO STATIC

Add one Brownian/Slow LFO route; create a contrasting Scene B; lengthen listening time.

TOO BRIGHT

Reduce Air/shimmer; choose a softer Character; lower bright FX feedback.

TOO MUDDY

Reduce Base/density; narrow Mass; simplify low-register voices; rebalance layers.

TOO OBVIOUSLY CYCLIC

Use longer LFO periods; combine different timescales; substitute Smooth Random/Brownian.

TOO RANDOM

Re-establish Key/Scale locks; lower Rare Events; replace chaos with slow deterministic motion.

TOO LOUD

Reduce density/Mass/FX feedback and output; compare bypassed level at sensible monitoring volume.

CPU SPIKES

Reduce voice/density complexity and expensive atmospheric processing; test long sessions in the target DAW.

Useful feedback makes Syndrone better

Report repeatable behaviour, not just the symptom

INCLUDE

DAW + version
OS + version
AU or VST3
Preset / Structure
Exact steps
Whether it happens after reload
Approximate CPU behaviour

WATCH FOR

Clicks or discontinuities
CPU spikes
State-recall errors
Unexpected gain changes
Preset/randomise problems
Long-session instability
Host-specific behaviour

REPRODUCTION TEMPLATE

1. Start from named preset.
2. Change listed controls.
3. Save/reload if relevant.
4. Describe expected result.
5. Describe actual result.
6. State whether it happens every time. Screenshots or a short audio example are useful when the issue is visual or sonic.

FEEDBACK > hello@slowfieldaudio.com

Quick-reference vocabulary

A compact translation of the concepts used throughout this guide.

STRUCTURE

Core synthesis architecture / voice-building behaviour.

CHARACTER

Macro tonal and behavioural voicing layered onto Structure.

SCENE

Captured musical state used as a morph destination.

EVOLUTION

Long-form change over a selected time period and shape.

SOURCE

A modulation signal that changes over time.

DESTINATION

The parameter controlled by a modulation source.

AMOUNT

Strength and direction of a modulation connection.

BROWNIAN / RANDOM WALK

Wandering stochastic motion suited to organic drift.

EUCLIDEAN PULSE

Evenly distributed events across a rhythmic cycle.

RARE EVENTS

Occasional generative deviations used as punctuation.

MASS

Perceived scale/density enhancement.

AIR

Uppermost layer in the four-band conceptual mixer.

Build -> animate -> spatialise -> curate

A repeatable method for almost any Syndrone patch.

1 CONSTRAIN

Choose Key, Scale and octave range. Decide fixed form or Perpetual.

2 IDENTIFY

Choose Structure + Character. Set voice density and core tone.

3 EVOLVE

Create A/B scenes. Pick time, shape and moderate Amount.

4 MODULATE

Add one source -> destination route. Then add only what the patch needs.

5 SPACE

Set static space first, then orbit/linked movement and atmosphere.

6 BALANCE

Use Base, Middle, Top and Air. Check Mass, Supermassive and FX energy. Leave headroom.

7 LISTEN LONG

Run for 10-20 minutes. Remove obvious repetition or excessive randomness. Save the successful state.

SYndrone is most convincing when its mechanisms disappear and the listener hears one slowly changing musical environment.

Slowfield Audio | Syndrone beta 1.0.5 | macOS + Windows | AU / VST3 | hello@slowfielddaudio.com